

MATH3018 Assignment 5, due on Mar 17

1. Recall the fact: if $Z \sim N_p(0, I_p)$ is independent of $V \sim W_p(n-1, I_p)$, then $Z^\top \left(\frac{V}{n-1}\right)^{-1} Z \sim \frac{(n-1)p}{n-p} F_{p, n-p}$. [2pts each]

(a) Suppose we have independent samples: $\mathbf{x}_1, \dots, \mathbf{x}_n \stackrel{\text{i.i.d.}}{\sim} N_p(\mu, \Sigma)$. Let $\bar{\mathbf{x}}$ be the sample mean and $\mathbf{S}$ be the sample covariance matrix. Show that for any $C \in \mathbb{R}^{m \times p}$ satisfying $C\Sigma C^\top$ is invertible, we have

$$n [C(\bar{\mathbf{x}} - \mu)]^\top [C\mathbf{S}C^\top]^{-1} [C(\bar{\mathbf{x}} - \mu)] \sim \frac{(n-1)m}{n-m} F_{m, n-m}$$

(b) Suppose we have samples from two independent populations: $\mathbf{x}_{11}, \dots, \mathbf{x}_{1n_1} \stackrel{\text{i.i.d.}}{\sim} N_p(\mu_1, \Sigma)$, $\mathbf{x}_{21}, \dots, \mathbf{x}_{2n_2} \stackrel{\text{i.i.d.}}{\sim} N_p(\mu_2, \Sigma)$. Show that

$$\begin{aligned} (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2 - (\mu_1 - \mu_2))^\top \left[\left(\frac{1}{n_1} + \frac{1}{n_2} \right) S_{\text{pooled}} \right]^{-1} (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2 - (\mu_1 - \mu_2)) \\ \sim \frac{(n_1 + n_2 - 2)p}{n_1 + n_2 - p - 1} F_{p, n_1 + n_2 - p - 1}, \end{aligned}$$

where $\bar{\mathbf{x}}_1$ and $\bar{\mathbf{x}}_2$ are the sample mean for $\{\mathbf{x}_{1i}\}$ and $\{\mathbf{x}_{2i}\}$.

Solution:

(a) First, recall two basic properties:

$$\begin{aligned} \sqrt{n}(\bar{\mathbf{x}} - \mu) &\sim N_p(0, \Sigma), \\ (n-1)\mathbf{S} &\sim W_p(n-1, \Sigma). \end{aligned}$$

Then for any $C \in \mathbb{R}^{m \times p}$, we have

$$\begin{aligned} \sqrt{n}(C\bar{\mathbf{x}} - C\mu) &\sim N_m(0, C\Sigma C^\top), \\ (n-1)C\mathbf{S}C^\top &\sim W_m(n-1, C\Sigma C^\top). \end{aligned}$$

Let $\Gamma = C\Sigma C^\top$, which is invertible by assumption. Let $Z := \Gamma^{-1/2} \sqrt{n}(C\bar{\mathbf{x}} - C\mu)$, and $V := (n-1)\Gamma^{-1/2}(C\mathbf{S}C^\top)\Gamma^{-1/2}$. Then $Z \sim N_m(0, I_m)$, $V \sim W_m(n-1, I_m)$ and they are independent.

Therefore,

$$\begin{aligned}
& n(C\bar{\mathbf{x}} - C\mu)^\top (CSC^\top)^{-1} (C\bar{\mathbf{x}} - C\mu) \\
&= \left(\Gamma^{-1/2} \sqrt{n} (C\bar{\mathbf{x}} - C\mu) \right)^\top \left(\frac{(n-1)\Gamma^{-1/2} (CSC^\top) \Gamma^{-1/2}}{n-1} \right)^{-1} \left(\Gamma^{-1/2} \sqrt{n} (C\bar{\mathbf{x}} - C\mu) \right) \\
&= Z^\top \left(\frac{V}{n-1} \right)^{-1} Z \\
&\sim \frac{(n-1)m}{n-m} F_{m, n-m}
\end{aligned}$$

where in the second last step we substitute our definitions of Z and V , and in the last step we apply the fact stated in the problem.

- (b) Since $\mathbf{x}_{1i} \stackrel{\text{i.i.d.}}{\sim} N_p(\mu_1, \Sigma)$ and $\mathbf{x}_{2i} \stackrel{\text{i.i.d.}}{\sim} N_p(\mu_2, \Sigma)$, the sample means have distributions:

$$\bar{\mathbf{x}}_1 \sim N_p\left(\mu_1, \frac{\Sigma}{n_1}\right) \quad \bar{\mathbf{x}}_2 \sim N_p\left(\mu_2, \frac{\Sigma}{n_2}\right)$$

Since the two samples are independent, we have

$$\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2 \sim N_p\left(\mu_1 - \mu_2, \left(\frac{1}{n_1} + \frac{1}{n_2}\right) \Sigma\right) \quad (1)$$

Hence,

$$\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2 - (\mu_1 - \mu_2) \sim N_p\left(\mathbf{0}, \left(\frac{1}{n_1} + \frac{1}{n_2}\right) \Sigma\right) \quad (2)$$

By properties of Wishart distributions,

$$(n_1 + n_2 - 2)S_{\text{pooled}} = (n_1 - 1)S_2 + (n_2 - 1)S_2 \sim W_p(n_1 + n_2 - 2, \Sigma)$$

which is independent of $\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2$.

Let $Z = \left(\left(\frac{1}{n_1} + \frac{1}{n_2}\right) \Sigma\right)^{-1/2} (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2 - (\mu_1 - \mu_2))$, and $V = \Sigma^{-1/2} (n_1 + n_2 - 2)S_{\text{pooled}}$. Then $Z \sim N_p(0, I_p)$, $V \sim W_p(n_1 + n_2 - 2, I_p)$ and they are independent.

Therefore

$$\begin{aligned}
& (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2 - (\mu_1 - \mu_2))^\top \left[\left(\frac{1}{n_1} + \frac{1}{n_2}\right) S_{\text{pooled}} \right]^{-1} (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2 - (\mu_1 - \mu_2)) \\
&= Z^\top \left(\frac{V}{n_1 + n_2 - 2} \right)^{-1} Z \\
&\sim \frac{(n_1 + n_2 - 2)p}{n_1 + n_2 - 1 - p} F_{p, n_1 + n_2 - 1 - p}.
\end{aligned}$$

2. Samples of sizes $n_1 = 45$ and $n_2 = 55$ were taken of Wisconsin homeowners with and without air conditioning, respectively (assume both populations are multivariate normal and have the same population covariance matrix). Two measurements of electrical usage (in kilowatt hours) were considered. The first is a measure of total on-peak consumption (X_1) during July, and the second is a measure of total off-peak consumption (X_2) during July. The resulting summary statistics are

$$\bar{\mathbf{x}}_1 = \begin{bmatrix} 204.4 \\ 556.6 \end{bmatrix}, \quad \mathbf{S}_1 = \begin{bmatrix} 13825.3 & 23823.4 \\ 23823.4 & 73107.4 \end{bmatrix}, \quad n_1 = 45$$

$$\bar{\mathbf{x}}_2 = \begin{bmatrix} 130.0 \\ 355.0 \end{bmatrix}, \quad \mathbf{S}_2 = \begin{bmatrix} 8632.0 & 19616.7 \\ 19616.7 & 55964.5 \end{bmatrix}, \quad n_2 = 55$$

- (a). What is the pooled estimate of the population covariance matrix? [1pt]
 (b). Test H_0 : The two populations have equal means at the significance level 0.05. [2pts]
 (c). Find 95% Bonferroni intervals for the differences in the mean components. [2pts]

Solution

(a)

$$\mathbf{S}_{\text{pooled}} = \frac{n_1 - 1}{n_1 + n_2 - 2} \mathbf{S}_1 + \frac{n_2 - 1}{n_1 + n_2 - 2} \mathbf{S}_2 = \begin{bmatrix} 10963.7 & 21505.5 \\ 21505.5 & 63661.3 \end{bmatrix}$$

(b)

$$T^2 = (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)^\top \left[\left(\frac{1}{n_1} + \frac{1}{n_2} \right) \mathbf{S}_{\text{pooled}} \right]^{-1} (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2) = 16.07$$

$$\frac{(n_1 + n_2 - 2)p}{n_1 + n_2 - p - 1} F_{p, n_1 + n_2 - p - 1}(\alpha) = \frac{98(2)}{97} F_{2, 97}(0.05) = (2.02)(3.1) = 6.26$$

Since $T^2 > 6.26$, we reject H_0 .

- (c) With $\boldsymbol{\mu}'_1 - \boldsymbol{\mu}'_2 = [\mu_{11} - \mu_{21}, \mu_{12} - \mu_{22}]$,
 the Bonferroni intervals are

$$\mu_{1i} - \mu_{2i} : (\bar{x}_{1i} - \bar{x}_{2i}) \pm t_{n_1 + n_2 - 2} \left(\frac{\alpha}{2p} \right) \sqrt{\left(\frac{1}{n_1} + \frac{1}{n_2} \right) (S_{\text{pooled}})_{ii}}.$$

By calculations, for $\mu_{11} - \mu_{21} : [26.50, 122.30]$; for $\mu_{12} - \mu_{22} : [86.17, 317.03]$.

3. Two variables are observed from 3 treatments. Sample data are as follows:

Treatment1 : (6, 7), (5, 9), (8, 6), (4, 9), (7, 9);

Treatment2 : (3, 3), (1, 6), (2, 3);

Treatment3 : (2, 3), (5, 1), (3, 1), (2, 3).

Some useful statistics are:

Treatment 1: $n_1 = 5, \bar{x}_1 = (6, 8)^T$, and

$$S_1 = \begin{pmatrix} 5/2 & -3/2 \\ -3/2 & 2 \end{pmatrix}$$

Treatment 2: $n_2 = 3, \bar{x}_2 = (2, 4)^T$, and

$$S_2 = \begin{pmatrix} 1 & -3/2 \\ -3/2 & 3 \end{pmatrix}$$

Treatment 3: $n_3 = 4, \bar{x}_3 = (3, 2)^T$, and

$$S_3 = \begin{pmatrix} 2 & -4/3 \\ -4/3 & 4/3 \end{pmatrix}$$

- (a) Construct the MANOVA table, like Page 30 of Chapter 5 (part 1) slides. (Do not need to include the Wilk's lambda statistic Λ^* in the table) [3pts]
- (b) Find the Wilk's lambda statistic, then test $H_0 : \tau_1 = \tau_2 = \tau_3 = 0$ (no treatment effect) at significance level $\alpha = 0.01$. [3pts]

Solution

(a)

The one-way MANOVA table

Source of variation	Sum of square matrix	Degrees of freedom
Treatment	$\mathbf{B} = \begin{pmatrix} 36 & 48 \\ 48 & 84 \end{pmatrix}$	$3 - 1 = 2$
Residual	$\mathbf{W} = \begin{pmatrix} 18 & -13 \\ -13 & 18 \end{pmatrix}$	$5 + 3 + 4 - 3 = 9$
Total (corrected)	$\mathbf{B} + \mathbf{W} = \begin{pmatrix} 54 & 35 \\ 35 & 102 \end{pmatrix}$	$12 - 1 = 11$

(b)

$$\Lambda^* = \frac{|\mathbf{W}|}{|\mathbf{W} + \mathbf{B}|} = \frac{155}{4283} = 0.0362.$$

For $p = 2$ and $g = 3$ the test statistic is

$$\left(\frac{\sum_{l=1}^g n_l - g - 1}{g - 1} \right) \left(\frac{1 - \sqrt{\Lambda^*}}{\sqrt{\Lambda^*}} \right) = \frac{12 - 2 - 2}{2} \times \frac{1 - \sqrt{0.0362}}{\sqrt{0.0362}} = 17.0266$$

For $\alpha = 0.01$ the critical value of $F_{2(g-1), 2(\sum_{l=1}^g n_l - g - 1)}(\alpha)$ is $F_{4,16}(0.01) = 4.7726$. Since $17.0266 > 4.7726$, we reject H_0 at significant level $\alpha = 0.01$ and conclude that at least one τ_l is not zero.